

Yerin Hong

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PROFESSIONAL SUMMARY

Full-stack engineer with 8+ years of experience building production systems across web, backend, and platform layers. Hands-on with TypeScript, Python, React/Next.js, and NestJS, with recent work spanning AI tutoring products, exam management workflows, admin tooling, and cloud-based services. Comfortable owning features end-to-end from schema design and API implementation to frontend UX and production support.

CORE SKILLS

TypeScript, JavaScript, Python, SQL · React, Next.js, NestJS, FastAPI, Laravel
REST API design, schema design, cloud services · AWS Lambda, EC2, S3, Terraform
OpenAI, Gemini, internal AI service integration · UML, modular development
C++, Objective-C, Flutter · Linux Kernel, Android TEE, ARM Trusted Firmware · Shell, Git, Gerrit, SVN

EXPERIENCE

HappyEduTech | Senior Full-Stack & AI Engineer (Freelance Contract)

Remote 07/2025 – Present

Skills: Next.js, NestJS, TypeScript, Python, FastAPI, PyTorch, AWS Lambda, EC2, S3, Terraform, SQL, REST API

- Built end-to-end exam workflows across admin and student-facing systems, covering creation, editing, online assignment, submission, result retrieval, and history across Next.js frontends and NestJS APIs.
- Delivered asynchronous AI question-generation features with batch-job schemas, UUID tracking, polling APIs, and admin-side loading and failure states for long-running generation tasks.
- Improved the AI tutoring product by refining prompt architecture, streaming and caching behaviour, subject-specific response handling, markdown rendering, and multimodal image input.
- Built internal operational tooling including ZIP upload, Excel-based bulk updates, LaTeX image generation, batch status tracking, and search filters across curriculum hierarchy, question type, and role-based visibility.

uTriper | Solution Architect & Frontend Engineer (Freelance Contract)

Remote 07/2024 – 01/2025

Skills: React, SQL, REST API, System Design

- Advised on database architecture and feature logic for a booking management platform across two contract periods.
- Refactored and modularised the React codebase to improve maintainability and delivery speed.
- Built new user-facing pages through structured component design and cleaner frontend architecture.

Arum 3D Solutions LTD | Software Engineer, IT & Development Team

Hersham, UK 06/2022 – 06/2024

Skills: React, Laravel, SQL, Figma, Flutter, REST API, gRPC, CI/CD, Android, iOS

- Developed and maintained ERP features across frontend and backend using Laravel and React, including a migration from Blade to React and a Laravel upgrade from v7 to v10.
- Designed REST APIs and improved route structure and performance for internal business systems.
- Built internal tooling and product workflows backed by relational database schema design, and developed Flutter-based mobile features with CI/CD automation support.

MediaTek | System Software Engineer, System Software Team

Seongnam, South Korea 03/2020 – 09/2021

Skills: C/C++, Shell, Python, Android, Linux Kernel, TEE, Secure Boot, ARM Trusted Firmware, Git, Gerrit

- Debugged Android TEE and secure boot components including RPMB and ARM Trusted Firmware; contributed to platform-level OS porting and validation of embedded security features.
- Supported Samsung and LG mobile projects with system-level technical solutions and security compliance validation.
- Migrated internal build configurations from Android.mk to Android.bp for platform consistency.

AhnLab | Software Engineer, V3 Team

Seongnam, South Korea 01/2018 – 03/2020

Skills: C++, Objective-C, Python, Flask, Redis, Docker, UML, SVN, Git, macOS

- Developed macOS antivirus components in C++ and Objective-C, covering UI and system binaries; applied UML design and modular development strategies to ensure maintainability.
- Built and maintained a virtualisation-based automation server using Python, Flask, Redis, and Docker for internal test infrastructure.
- Promoted from Junior Software Engineer to Software Engineer in January 2020.

Naver Labs | Intern, Robotics Group

Seongnam, South Korea 07/2016 – 08/2016

Skills: C/C++, MFC, Windows, Serial Communication, UWB

- Built a relative positioning system using MFC and UWB hardware (EVB1000-based), and implemented a Kalman filter to enhance accuracy and stability of location estimation.

RESEARCH EXPERIENCE

Explainable LLM-based Chess Agent (AlphaChess) — AIFEL, MODULABS | 04/2025 – 06/2025

- Researched how CoT-based SFT and GRPO improve both performance and explainability in LLM-based chess agents.
- Designed experiments using prompt engineering, multi-agent prompting, and RAG to enhance strategic reasoning.
- Project link: [\[link — add URL\]](#)

Interpretable Melanoma Diagnosis with Pseudo-Depth Estimation — | 06/2025

- Investigated a Mamba2D-based pseudo-depth estimation method to interpret melanoma predictions through relative depth cues and Grad-CAM-style visualisations.

EDUCATION

AIFEL, MODULABS | Seoul, South Korea

AI Research Programme | 12/2024 – 07/2025

- Studied deep learning fundamentals, paper implementation, and real-world AI applications.
- Participated in a DLthon project classifying threatening conversations using Korean pretrained models.

Kwangwoon University | Seoul, South Korea

M.S., Electronics and Communications Engineering | 03/2016 – 02/2018

- Digital Signal Processing Lab, supervised by Prof. Hyukjun Oh.
- Visiting Student, Prof. Yung Yi's Lab, KAIST — 08/2017 – 09/2017.
- Coursework: Medical Image Processing, Introduction to Machine Learning, Advanced Image Processing, Bio Signal Processing, Digital Image Surveillance System Design.

Kwangwoon University | Seoul, South Korea

B.S., Electronics and Communications Engineering | 03/2012 – 02/2016

- Embedded System Lab, supervised by Prof. Hyunseok Lee.
- Coursework: Mobile & Wireless Communication, Digital Communication, Image Signal Processing, Embedded Systems, Computer Architecture, Operating Systems, Digital Signal Processing.

PUBLICATIONS & PATENT

Y. Hong, G. Lee, and H. Oh, "Compare of channel coding for 5G system: LDPC and polar code," ICEIC, 2018. [\[link\]](#)

Y. Hong and H. Oh, "Improving the frequency domain resolution of wireless signal for observing the Doppler frequency," JKIIICE, vol. 21, no. 12, pp. 2270–2278, Dec. 2017. [\[link\]](#)

Y. Hong, H. Oh, and S. H. Won, "Method for subscriber authentication in cellular IoT device," U.S. Patent 9,888,385, Feb. 6, 2018. [\[link\]](#)

REFERENCES

Available upon request.